Project Initialization and Planning Phase

Date	29 June 2026
Team ID	SWTID-2026-7290
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview

Objective

The primary objective of Smart Lender is to automate and improve the loan approval process by using machine learning techniques to predict the creditworthiness of loan applicants. The system enables banks to make faster, more accurate, and data-driven lending decisions while reducing manual effort and minimizing the risk of loan defaults.

Scope

The project focuses on developing a web-based loan approval prediction system that:

- Collects applicant information through a user-friendly web interface.
- Preprocesses applicant data for machine learning prediction.
- Compares multiple classification algorithms:
  - Decision Tree
  - Random Forest
  - K-Nearest Neighbors (KNN)
  - XGBoost
- Selects the best-performing model (XGBoost).
- Predicts whether a loan application is likely to be approved.
- Deploys the solution on IBM Cloud for secure and scalable access.

Problem Statement

Description

Banks receive a large number of loan applications every day. Traditional manual evaluation is time-consuming, inconsistent, and susceptible to human error, resulting in delayed approvals and increased financial risk. There is a need for an intelligent system that automates loan eligibility assessment using historical applicant data.

Impact

Implementing Smart Lender will:

- Reduce loan processing time.
- Improve prediction accuracy.
- Minimize non-performing assets (NPAs).
- Increase operational efficiency.

Improve customer satisfaction through faster decisions.

Proposed Solution

Approach

Smart Lender uses supervised machine learning to analyze applicant information such as income, employment status, education, credit history, loan amount, and property area. Multiple classification models are trained and evaluated, with XGBoost selected as the final model due to its superior performance. The trained model is integrated into a Flask web application for real-time loan approval prediction.

Key Features

- Machine learning-based loan approval prediction.
- Comparison of multiple classification algorithms.
- XGBoost model with high prediction accuracy.
- Real-time prediction through a Flask web interface.
- Secure and responsive user interface.
- IBM Cloud deployment.

Easy integration with banking workflows.

Resource Requirements

Hardware

Resource Type	Description	Specification
Processor	Computing Resources	Intel Core i5 or higher
Memory	RAM	Minimum 8 GB
Storage	Disk Space	256 GB SSD or higher

Software

Resource Type	Description	Specification
Programming Language	Development	Python 3.x
Framework	Backend	Flask
Machine Learning Libraries	Model Development	Scikit-learn, XGBoost
Data Processing	Data Analysis	Pandas, NumPy
Model Storage	Serialization	Joblib
IDE	Development Environment	Jupyter Notebook, Visual Studio Code

Dataset

Resource Type	Description	Specification
Dataset	Loan Prediction Dataset	Kaggle Loan Prediction Dataset (CSV format, approximately 614 records)